Low-Energy Neutrino Event Reconstruction in DUNE: Clustering, Vertexing, and Energy Reconstruction

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Abstract

The Deep Underground Neutrino Experiment (DUNE) aims to characterize the two most energetic components of the low–energy flux of solar neutrinos (^{8}B & *hep*), as well as neutrinos from other astrophysical sources such as nearby supernovae or the diffuse supernovae neutrino background. The sensitivity to the charged current channel, with orders of magnitude higher cross–section than the elastic scattering interaction, together with DUNE's high exposure and great energy resolution, present an exceptional advantage in understanding the sharp energy characteristics of this neutrino flux. This work presents a detailed study of low–energy charged-current neutrino event reconstruction and energy resolution performance in the presence of significant radiological and external backgrounds. We describe the development and validation of a clustering algorithm that identifies primary ionization signals and associates adjacent energy depositions to account for spatially correlated activity. A dedicated energy calibration strategy is employed to recover the interacting neutrino energy. The impact of various background scenarios on the reconstruction is assessed, and mitigation strategies are discussed. We report on the energy resolution achieved across a range of true neutrino energies, highlighting the algorithm's effectiveness in maintaining reconstruction fidelity in the low-energy regime. These results demonstrate the robustness of the reconstruction approach and its potential contribution to the broader low–energy DUNE physics program.

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1 Introduction

There is a long history of advancements in particle and astroparticle physics resulting from measuring astrophysical neutrinos with energies in the 10 MeV range [1, 2]. The sun produces a continuous flux of roughly $7 \times 10^{10}\,\mathrm{s}^{-1}\,\mathrm{cm}^{-2}$ neutrinos as a byproduct of fusion reactions within the stellar core [3]. Despite the confirmation of solar neutrino oscillations as a solution to the *Solar Neutrino Problem* [4], several questions remain [5]. The Deep Underground Neutrino Experiment, as next–generation large–scale neutrino detector, will have the ability to shed light on some of them.

1.1 DUNE Far Detectors

DUNE will deploy four Far Detector (FD) modules in the Sanford Underground Research Facility (SURF) to survey a wide portfolio of physics topics. The FDs will be built in two phases. The first stage will include two Liquid Argon Time Projection Chamber (LArTPC)s with approximately 14 kt of active volume, which will be based on two distinct geometries, a Horizontal–Drift (HD) and a Vertical–Drift (VD) design. The HD's fiducial volume will extend 14.4 m in the drift direction, aligned along the horizontal coordinate x, 12 m along the vertical coordinate y, and 58 m along the remaining z coordinate. This module will be separated into four distinct drift volumes along the drift direction x. Alternating three Anode Plane Assembly (APA) and two Cathode Plane Assembly (CPA), which produce a drift potential of 180 kV, giving an electric field of $500\,\mathrm{V\,cm}^{-1}$.The FD VD module simplifies the design by dividing the active mass into just two volumes of dimensions $13.5 \times 6.5 \times 60\mathrm{m}^3$ separated horizontally by the cathode plane. Both detectors incorporate a **photon detection system** (PDS) which makes use of X–ARAPUCA (XA) tiles [6] distributed over the APA (for HD) and cathode and membrane planes (for VD) respectively.

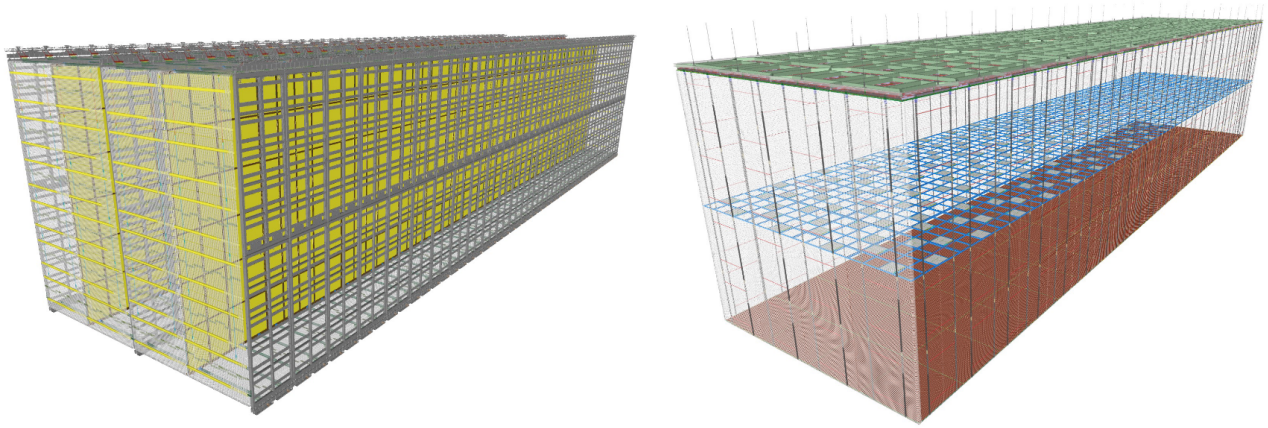


Figure 1: (left) DUNE FD HD module. (right) DUNE FD VD module.

1.2 Background Model

The measurement of low–energy neutrinos (<20 MeV), particularly those from the Solar ^{8}B chain, requires a thorough understanding of intrinsic (emitted from the Liquid Argon (LAr) or detector components) and external (penetrating from the cavern) backgrounds. One of the most studied backgrounds for LAr detectors is Radon (^{222}Rn), which is a noble gas produced in the decay chain of ^{238}U [7, 8]. As a gas, it can diffuse into the active volume of LAr detectors, where its decay daughters may plate out on surfaces or remain suspended, producing problematic beta and gamma radiation. Due to its nature as a noble gas, radon is difficult to remove from the detector environment, making it a persistent background source also in detector components. Among its progeny, several isotopes are particularly important for background modelling:

- **Bismuth (^{214}Bi)**: A short–lived beta emitter with a high Q–value (up to 3.27 MeV) and multiple gamma lines, making it the most significant source of high–energy background [9].
- **Bismuth (^{210}Bi)**: A medium–energy beta emitter contributing a continuum of background up to 1.16 MeV.
- **Lead (^{210}Pb)**: A long–lived isotope ($t_{1/2} = 22.3$ years) that accumulates on detector surfaces, acting as a persistent low–energy background source [10].
- **Polonium (^{210}Po)**: An alpha emitter with a half–life of 138 days, often used as a surface contamination proxy due to its distinct alpha signal [11]. Alpha particles have a very short range in LAr, but due to their

high energy and quenching effects, they can produce significant light signals in the detector, which can be misidentified as low–energy neutrino interaction candidates in the Photon Detection System (PDS).

Understanding the contributions from these (and other) isotopes is crucial in achieving a correct vertex and energy reconstruction required for low–energy neutrino analyses. Apart from the previously mentioned radon chain, a comprehensive list of background sources has been implemented in DUNE's simulation. These backgrounds can be classified into three main categories: intrinsic backgrounds, radiological or component–induced; and external backgrounds from the cavern. Evaluating the full scope of DUNE's background model is out of the scope of this work. Here we focus on giving a brief summary of the main component and discussing the features that have the biggest impact on the low–energy neutrino reconstruction.

- **External Backgrounds**: External radiation originates outside the detector cryostat. The flux is divided into two kinds of particles that manage to penetrate into the detector:

  1. **Cavern Neutrons**: Produced by (α, n) reactions in surrounding rock or via cosmic–ray spallation; they can enter the active volume and undergo elastic scattering or capture, producing a gamma–cascade which can mimic the topology of Charged Current (CC) solar neutrino interactions. Some neutrons are captured on the cryostat components before entering the LAr, contributing to the absolute environmental gamma flux.
  2. **Environmental gammas**: High–energy photons from the rock or shotcrete surrounding the detector, typically originating from natural radioactivity. The highest rate contributor to this flux in DUNE's caver at SURF is the shotcrete covering the walls. Low–energy gammas (<2.7 MeV) entering the active volume have a huge flux contributing mostly to worsen the signal reconstruction by mimicking the de–excitation gammas of the CC interaction or making the topological distinction between Elastic Scattering (ES) and CC less reliable. The higher energy external gammas (2.7 < E <14 MeV), produced by γ–decay of isotopes like ^{208}Tl (∼3.5 MeV) or ^{214}Bi (∼3.27 MeV) and secondary nuclear reactions of (n, γ) interactions in certain geological environments can produce gammas >6 MeV. These high–energy gamma rays are highly site-specific, depending on the local neutron flux and rock composition, therefore in situ measurements are fundamental to assess their impact on the detector.

- **Component-Induced Backgrounds**: Backgrounds can arise from the materials used to construct the detector itself. Material screening and selection, as well as radon mitigation systems, are essential to reduce these sources. These include decay chains of long–lived isotopes introduced during fabrication or operation. Some of the most prominent are:

  1. **Uranium (^{238}U)**: Produces alpha, beta, and gamma emissions, as well as daughters like ^{214}Bi and ^{210}Pb.
  2. **Thorium (^{232}Th)**: Similar to the uranium chain, with prominent gamma emitters like ^{208}Tl (2.6 MeV).
  3. **Potassium (^{40}K)**: Often present in glass, Photomultiplier Tube (PMT)s, and plastics; emits a 1.46 MeV gamma ray.
  4. **Radon (^{222}Rn)**: Present in atmospheric air and therefore very difficult to prevent.

- **Intrinsic Liquid Argon Backgrounds**: Certain radioactive isotopes are either present naturally in atmospheric argon or may be introduced as impurities. These form the intrinsic background component of LAr:

  1. **Argon (^{39}Ar)**: A beta emitter with an endpoint energy of 565 keV and a half–life of 269 years. It dominates the low–energy spectrum in atmospheric argon with an activity of $\sim 1\,\mathrm{Bq\,kg^{-1}}$. Together with the external gammas, this background contributes to the smearing of signals and worsens the CC and ES separation.
  2. **Argon (^{42}Ar) and Potassium(^{42}K)**: Rare isotopes that decay via beta emission. ^{42}K emits an electron ($Q \sim 3.5$MeV), which is particularly problematic due to its similarity to low–energy solar neutrino events.
  3. **Krypton (^{85}Kr)**: A beta emitter with a Q-value of 687 keV, introduced via contamination from air or during purification. Often requires cryogenic distillation to remove.
  4. **Radon (^{222}Rn)**: In addition to emanation from components, radon can dissolve in liquid argon.

In DUNE, LAr purification and recirculation installations are planned to ensure constant operating conditions. A detailed background model informed by in situ measurements, simulations, and radiopurity assays has been developed and is under constant revision to ensure accurate predictions of the background fluxes in the LAr active

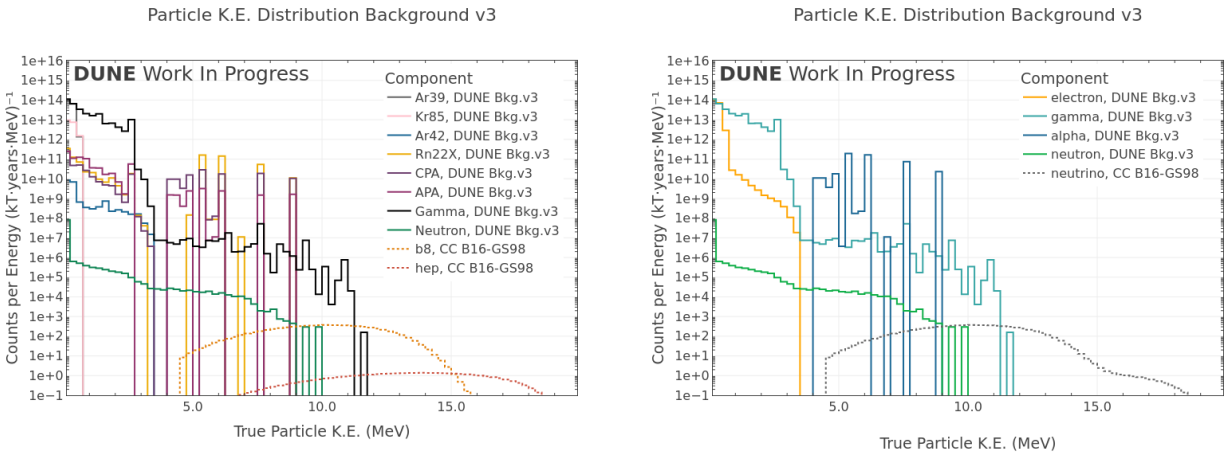


Figure 2: Simulated particle flux from background model v.3 on top of predicted solar spectrum. (left) Divided by the background component. (right) Divided by particle type.

volume. The model is implemented in Liquid Argon Software Framework (LArSoft), the software framework used for DUNE simulations, and is used to generate realistic event samples for detector response studies and physics analyses.

In fig. 2, we show the simulated particle flux from the background model v.3 on top of the predicted solar spectrum. The left panel shows the flux divided by background components, while the right panel shows it divided by particle type. The flux dominated by external gammas, argon and krypton at low– and external gammas and neutrons at high–energies. The figure highlights the importance of understanding these backgrounds by highlighting the huge difference in flux between the low–energy neutrino signal and the background components. Without a proper energy calibration, the smearing of the background components can lead to a significant loss of signal efficiency and a degradation of the energy resolution, which is crucial for low–energy neutrino analyses.

2 Event Simulation

In a LArTPC, charged particles ionize the argon atoms and an applied electric field causes the liberated electrons to drift toward segmented readout planes, where their arrival times and charge are recorded, allowing for a 3D–reconstruction of the interaction $< t, y, z >$. In addition to ionization signals, LAr also produces scintillation light, which can be used for triggering and timing (reconstructing the drift coordinate x). For low–energy events, efficient charge collection and light detection are crucial for a reliable reconstruction. The neutrino event simulation in DUNE follows the standard low–energy workflows of the Monte Carlo (MC) framework LArSoft for DUNE [12].

2.1 Geometries

In this work, we utilized two geometries: the HD workspace geometry (hd_1x2x6) and the VD workspace geometry (vd_1x8x14_3view_30deg). The HD workspace features a single central APA and a drift length of 3.6 m, designed to represent the central region of the actual HD detector. In contrast, the VD workspace geometry consists of a single drift volume that is approximately 7 m long (see fig. 4). Both geometries aim to model the full DUNE Far Detector (FD) modules by representing a sub–volume that can be scaled to the entire detector

The HD geometry is used in low–energy neutrino studies, simulating two background scenarios: one with complete external background (hd_1x2x6_lateralAPA) and another without background in the central APA (hd_1x2x6_centralAPA). In both scenarios, we simulate the full set of radiological backgrounds, along with top, bottom, head, and tail external backgrounds.

For the VD geometry, we consider two scenarios: the nominal scenario, which likely represents the upper half of the VD module (vd_1x8x14_3view_30deg), and a shielded scenario that assumes additional reduction of backgrounds in the form of altered fluxes (vd_1x8x14_3view_30deg_shielded). The latter scenario is likely to represent the lower half of the VD module, where shielding is more suitable from and engineering point of view.

2.2 Neutrino Interaction

The first step in simulating neutrino interactions in DUNE relies on a comprehensive nuclear model of the possible processes. For low–energy events, this is Model of Argon Reaction Low Energy Yields (MARLEY), an event

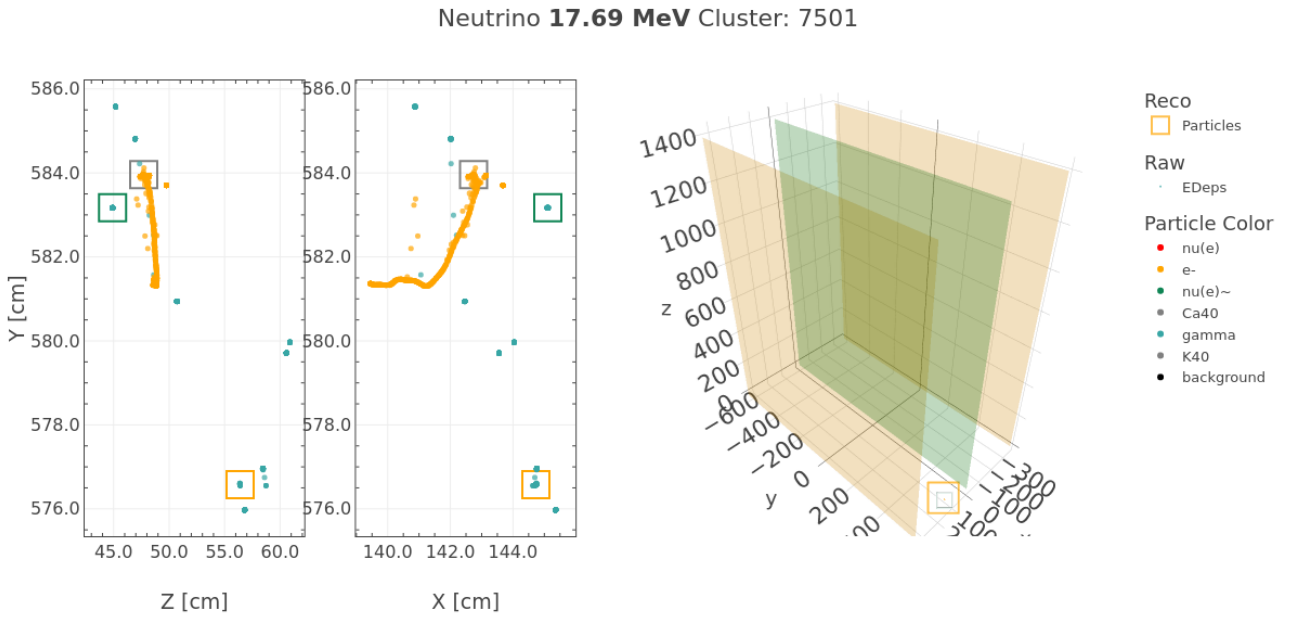


Figure 3: Example of a simulated ∼18 MeV neutrino interaction in the HD geometry. (left) The event displays the two Time Projection Chamber (TPC) projections of the primary electron energy depositions (orange) and the associated de–excitation gammas (turquoise). The energy deposits from the truth information of the signal propagation are marked with dots and the reconstructed clusters with empty squares. (right) The reco objects are displayed in a 3D representation of the workspace geometry. The represented planes are the central APA (green) and the corresponding CPAs (orange)

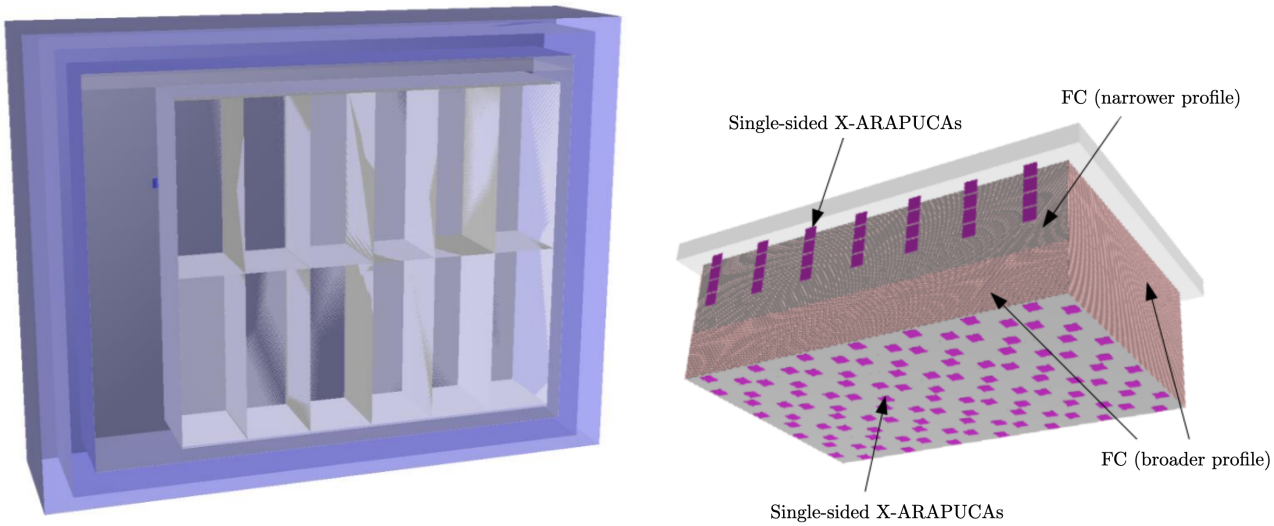


Figure 4: (left) The HD workspace geometry (hd_1x2x6_centralAPA) with a single central APA and a drift length of 3.6 m. (right) The VD workspace geometry (vd_1x8x14_3view_30deg).

generator developed to simulate the interactions of low–energy neutrinos in the range of a few to tens of MeV with argon nuclei. It is particularly suited for simulating interactions relevant to solar and supernova neutrinos in liquid argon detectors. In this work we focus on the CC process (although ES and Neutral Current (NC) have also benn implemented):

$$\nu_e + {}^{40}\mathrm{Ar} \rightarrow e^- + {}^{40}\mathrm{K}^* \tag{1}$$

The excitation of the residual nucleus ${}^{40}\mathrm{K}^*$ leads to the emission of de–excitation γ–rays, which can produce observable signals (blips) in the vicinity (<1 m) of the main electron (track). MARLEY incorporates gamma–decay schemes to describe these processes and feeds the resulting daughter particles into LArSoft, which is used to produce MC samples for physics studies. It simulates primary interaction kinematics, residual nuclear excited states and the gamma-ray cascades from nuclear de–excitation, as well as, other secondary processes.

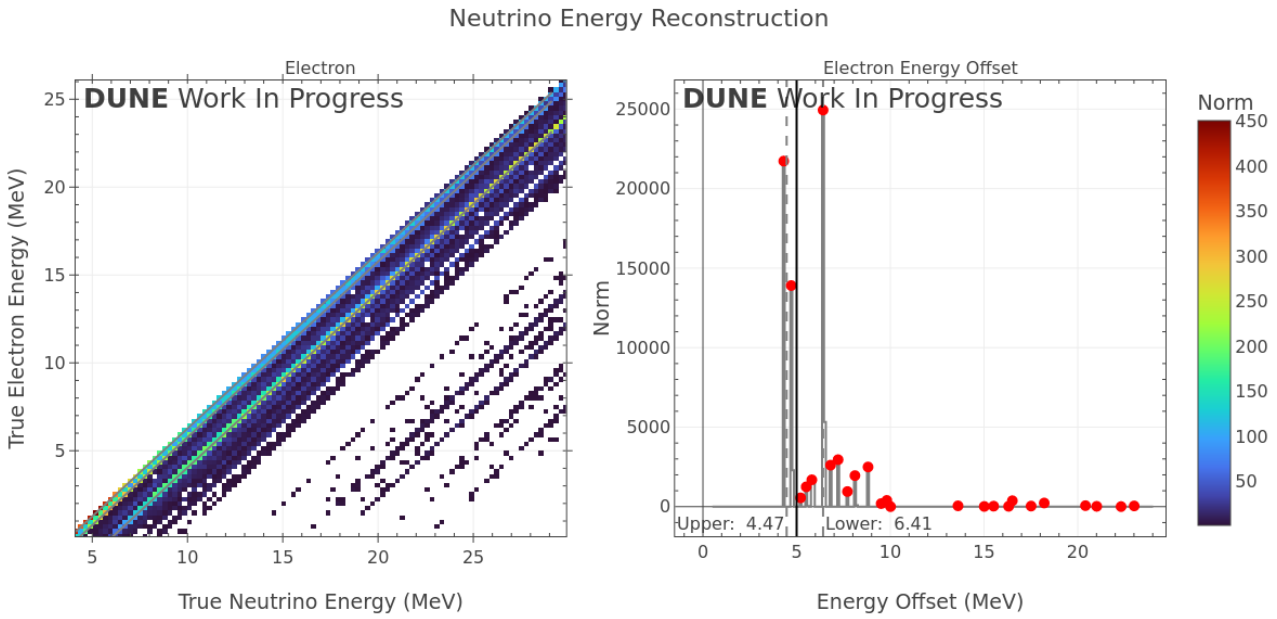


Figure 5: Neutrino–electron energy correlation in a CC interaction as predicted by MARLEY.

In fig. 6, the average energy transfer and binding energy of the CC interaction are displayed. One can observe how the gamma channel starts dominating the interaction up to ∼10 MeV, and above that threshold, the main electron takes over the fractional energy. This trend is accompanied by a continuous increase in average kinematic threshold caused by the production of additional secondary particles. The most important secondary production is the emission of neutrons, which happens above ∼15 MeV, and leads to a decrease in visible energy that cannot be recovered under realistic reconstruction scenarios (with background). All of these effects combine to a strong energy dependence of the deposited and visible energies, which should be taken into account in our signal calibration. Given that the true neutrino energy reconstruction is our calibration goal, the here presented calorimetric algorithms use the number of hits in the main clusters as a proxy for neutrino energy to compensate for the different physics of each energy range. The following electron 3.4 and neutrino calibration 3.6 steps divide the sample in number of clustered hits.

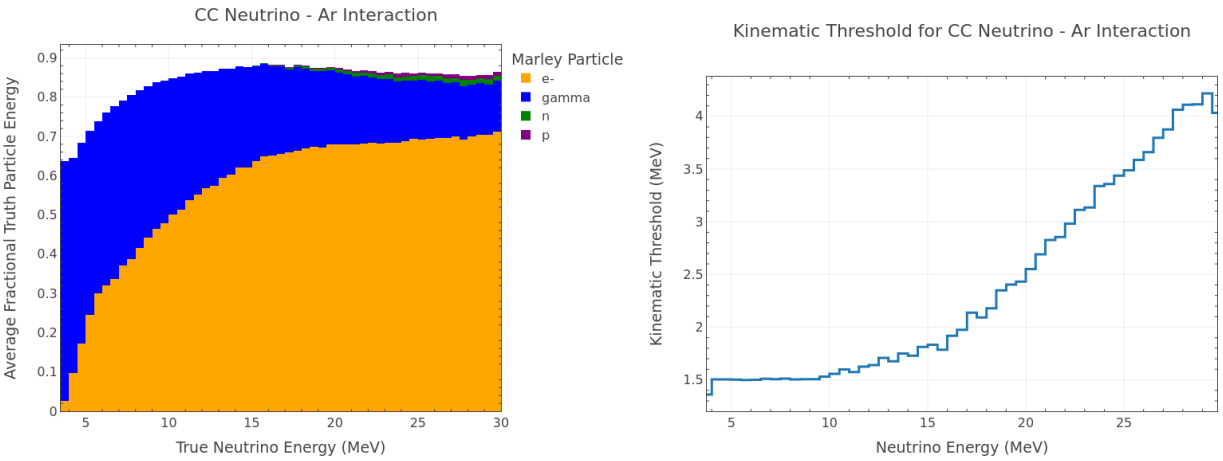


Figure 6: MARLEY's computation of CC neutrino interaction. (left) Daughter fraction of the original kinetic energy. (right) Average kinematic threshold.

It is important to mention at this point that a typical low–energy MC production include a full propagation of the backgrounds described in section 1.2. These additional particles are produced in their specific rates and concentrations according to computations that are out of the scope of this publication. All particle kinematics and detector placement happen in the so–called *gen* stage of the simulation. All produced background particles follow the same workflow chain as the signal particles and contribute to the ultimate smearing and resolution of the reconstructed objects.

2.3 Particle Propagation

Following the generation of the primary neutrino interaction via the MARLEY generator for low–energy solar neutrino events, the resulting final–state particles are propagated through the liquid argon medium using a Geant4 simulation. This *g4* stage, models the detailed particle–matter interactions, including energy loss via ionisation, multiple scattering, bremsstrahlung, photon production, and hadronic interactions, according to physics lists tailored for low–energy processes.

As charged particles traverse the liquid argon, they lose energy primarily through ionization, producing free ionization electrons and excitation of argon atoms, which subsequently decay and emit prompt scintillation photons in the vacuum ultraviolet (VUV) range (128 nm). These processes are central to the signal generation in a LArTPC. For computational efficiency, the simulation does not explicitly track every ionisation electron or VUV photon. Instead, energy loss along particle trajectories is recorded as a set of energy depositions, or `edeps`, which include position, time, energy, and associated particle identity. Similarly, the number of scintillation photons produced in each energy deposition step is recorded as a secondary output, commonly referred to as `simphotons`. These `edeps` and `simphotons` are then effectively propagated through the LAr volume with custom analytical algorithms in a subsequent stage of simulation, often called *detsim*.

2.4 Detector Simulation

In this stage, after the effective propagation of ionisation electrons and scintillation light to the TPC (wires or strips) and PDS (photodetectors) readout elements, the *detsim* step models the signal formation on the readout systems. This stage takes as input the energy depositions (`edeps`) and the associated scintillation photons (`simphotons`) from the particle propagation step, and simulates the detector response to these inputs, producing digitised signals based on the detector's readout electronics and noise characteristics. The two components of the detector response simulation are:

2.4.1 TPC Signal

Ionisation electrons drift under an applied electric field toward the TPC wire planes. During this drift, the electrons experience diffusion (both longitudinal and transverse), attachment to electronegative impurities (parametrised by the electron lifetime), and charge loss due to recombination. These effects are modelled to modify the total collected charge and the temporal and spatial spread of the signal. DUNEs LArTPC use three wire planes (also called "views") to detect the drifting charge (1 collection plane and 2 induction planes). The wire planes are oriented at different angles to allow for 3D reconstruction of the ionisation signal.

1. **Collection plane** responds with unipolar signals as the electrons are finally absorbed.
2. **Induction planes** respond with bipolar signals as electrons pass by without being collected.

The motion of charge induces time-dependent signals (waveforms) on the wires, which are digitised at high frequency. These raw waveforms are then processed using simulated electronic noise based on detector-specific measurements, which is applied to emulate realistic signal-to-noise conditions and electronics shaping. This full set of operations are performed in LArSoft using the Wire–Cell toolkit [13]. To improve charge extraction, the waveforms undergo a **deconvolution procedure**, where the electronics response and noise spectrum are inverted in frequency space to recover a cleaner signal representation. The deconvolved waveforms are integrated over time windows to form localised `hits`, which are stored with time, position, and charge–profile information.

2.4.2 PDS Signal

Simultaneously, the scintillation photons are detected by the PDS, composed of XAs. The `simphotons` from Geant4 are converted into Photoelectron (PE)s using detector-specific quantum efficiencies and timing models.

Similarly to the TPC reconstruction, the PDS response is also emulated with realistic timing resolution, dark noise, and single photoelectron gain characteristics. Simulated waveforms are produced for each optical channel, overlaid with noise, and optionally deconvolved or filtered. The resulting waveforms are processed to extract *optical hits*(integrated photon pulses) with associated time and PE information.

2.5 Object Reconstruction

As a final step, after the low–energy simulation steps of generation, propagation and detector simulation of the MC workflow. The reconstruction step for these studies includes a basic clustering scheme that applies both to the TPC and PDS hits generating the final TPC clusters and PDS flashes to be used in the analysis. This *reco* stage typically is subdivided into two sub–steps:

1. *Reco1*: The TPC `hits` are grouped according to tight space (by selecting a range of adjacent channels) and time constrains to form `clusters`, which aim to include all the charge from localized ionisation signals of single particles. The PDS `ophits` are grouped following the same logic but with a much broader space range to account for the larger spread of scintillation light. The TPC and PDS code for this stage are implemented in the SolarCluster_module.cc and SolarOpFlash_module.cc files respectively.

2. *Reco2*: The *Reco1* TPC `cluster` and further grouped into `lowe events` containing one `main` and any number of found `adjacent clusters`. For each `main cluster`, a flash–matching algorithm is applied to select a single PDS `flash` object to reconstruct the drift–coordinate of the event.

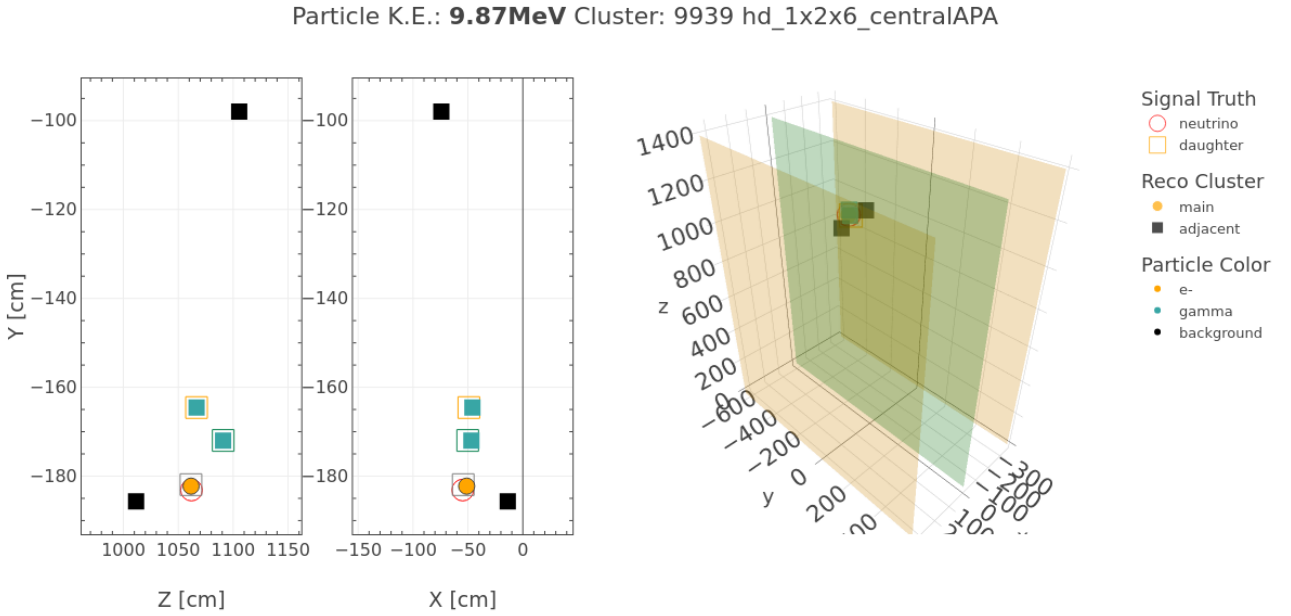


Figure 7: Example of a simulated ∼10 MeV neutrino interaction in the HD geometry. (left) The event displays the two TPC projections of the primary electron cluster (orange), the associated de–excitation gammas (turquoise) and uncorrelated background activity (black). The biggest energy deposits from the truth information of the signal propagation are marked with empty squares and the reconstructed clusters with filled squares. (right) The same points are displayed in a 3D representation of the workspace geometry. The represented planes are the central APA (green) and the corresponding CPAs (orange)

An example of a simulated neutrino interaction in the HD geometry is shown in fig. 7. The event displays the two TPC projections of the primary electron cluster (orange), the associated de–excitation gammas (turquoise) and uncorrelated background activity (black). The biggest energy deposits from the truth information of the signal propagation are marked with empty squares and the reconstructed clusters with filled squares. Compared to fig. 3, the clusters are now reduced to a point–like data structure, which is the result of the clustering algorithm.

2.6 Event Analysis

For the specific purpose of the solar neutrino studies an analyzer module (SolarNuAna_module.cc) has been written to combine these reconstruction objects with the truth information of the event. This module essentially combines all the information needed to perform the calibration and reconstruction of the signal candidates. The biggest caveat from the analysis point of view is that all clusters and flash information is reduced to a point–like data structure. This is, of course, a choice of design in the analyzer and allows for a simpler handling of the data because all event candidates have the same dimension. In future works, this analyzer should be substituted by the more general LowEAna_module.cc which aims to provide the same event reconstruction while keeping the internal structure of the candidate clusters to allow for better calibration and discrimination strategies.

3 Neutrino Reconstruction Strategy

The calibration and characterization of the CC neutrino signal are based on the true information of the main electron of the interaction. If not otherwise mentioned, the plots in this section are produced from a MARLEY generated CC neutrino sample with central APA background. The first step of the reconstruction process

happens already when clustering the hits of the TPC and PDS detector subsystems in the simulation's reco1 stage (see section 3.1). Next, we evaluate the detection efficiency and completeness of the collection– induction–matched TPC clusters (see section 3.2). Subsequently, the event electron is corrected in charge for the drift time attenuation before computing a calibration factor that converts charge into deposited energy (see section 3.4). The calibration factor is computed as a function of the threshold in the cluster's number of hits (`nhits`), serving as a proxy for the electron energy. Finally, the calibration factor is applied to all adjacent clusters of the event candidate and the total neutrino energy is computed with four different algorithms (see section 3.6). During the TPC calibration process, the drift coordinate is assumed from truth and the factors are extracted in absence of background. A discussion on how the intrinsic background contamination of the detector affects the signal reconstruction at these low energies is shown in section 3.7 and the effects on drift coordinate resolution (as a result of TPC–PDS match) are presented in section 3.8.

3.1 Event Definition

Due to the lack of a higher–order reconstruction object in the standard low–energy workflow, DUNE's solar analysis starts with calibrating the clustered hits detected for the CC interaction simulated by the MARLEY generator. Each cluster is a point–like object characterised by its charge (summed over the charge of the individual hits) and a 3D–reconstructed vertex $< t, y, z >$ (computed from the weighted average of the individual hits). These clusters are divided into primary (with charge bigger than any other's cluster charge up to 1 m distance) or adjacent. We define a primary cluster together with its adjacent clusters and one possible associated flash as an event. Clustering algorithms have been developed (duneopdet & dunereco) with associated producer modules (SolarOpFlash_module.cc & SolarCluster_module.cc) and included in the solar_reco_dune10kt_1x2x6.fcl & solar_reco_dunevd10kt_1x8x14_3view_30deg.fcl standard reconstruction workflows.

3.2 Detection Efficiency

The first figure of merit to be extracted from the simulation is the detection efficiency (see fig. 8 (left)). We observe a strong dependence on the number of hits. Only with a minimum hit condition of 2, the simulation reaches detection efficiency values of ∼100%. Achieving a reconstruction threshold (>90% efficiency) above 5.5 MeV requires a minimum hit condition of 1 hit. Instead of interpreting this efficiency in terms of the neutrino electron, using the information provided in fig. 5 (neutrino–electron energy offset >4.5 MeV) one can deduce an electron detection threshold of ∼1 MeV. A better estimation of the detection threshold can be extracted from the energy resolution fit in fig. 14.

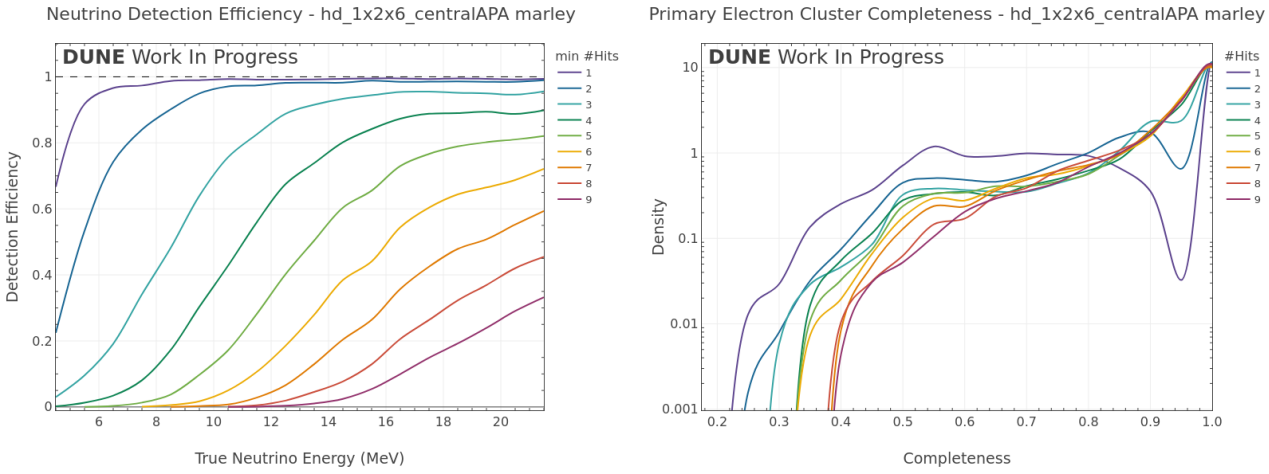


Figure 8: Efficiency plots for signal clustering parameters of 3 adjacent channels and 25 ticks. (left) Neutrino Detection Efficiency. (right) Main electron charge completeness in the primary cluster.

The next parameter to evaluate is the completeness of the primary clusters, which is a measure of the electron charge–confinement given a certain clustering parameter set. For the purpose of the solar neutrino analysis of DUNE, clustering parameters of 3 adjacent channels and 25 ticks have been used to maximize completeness while ensuring cluster purity in the context of background. The most up–to–date values for this clustering should be found in the producer's SolarCluster.fcl. Despite the good overall completeness of the signal, we observe a trend with respect to the number of hits per cluster, which can be accounted for in the later described charged correction factor to improve reconstruction. The trend that one observes is clear and understandable; for depositions which result in a lower number of hits (this can happen because of low energy deposition or a perpendicular trajectory

of the particle with respect to the TPC plane) it is more likely that clusters of similar sizes are formed (e.g. 2 clusters of 1 hit) resulting in a lower charge in the primary cluster.

3.3 Electron–Lifetime Correction

TPC hits are first grouped into spatial clusters using a dedicated clustering algorithm. Each event contains a **main cluster**, corresponding to the primary outgoing electron, and zero or more **adjacent clusters**, interpreted as gamma-induced "blips". The first step in calibration involves correcting the total ionization charge of each cluster for the electron–lifetime attenuation in the LAr (see section 3.3).

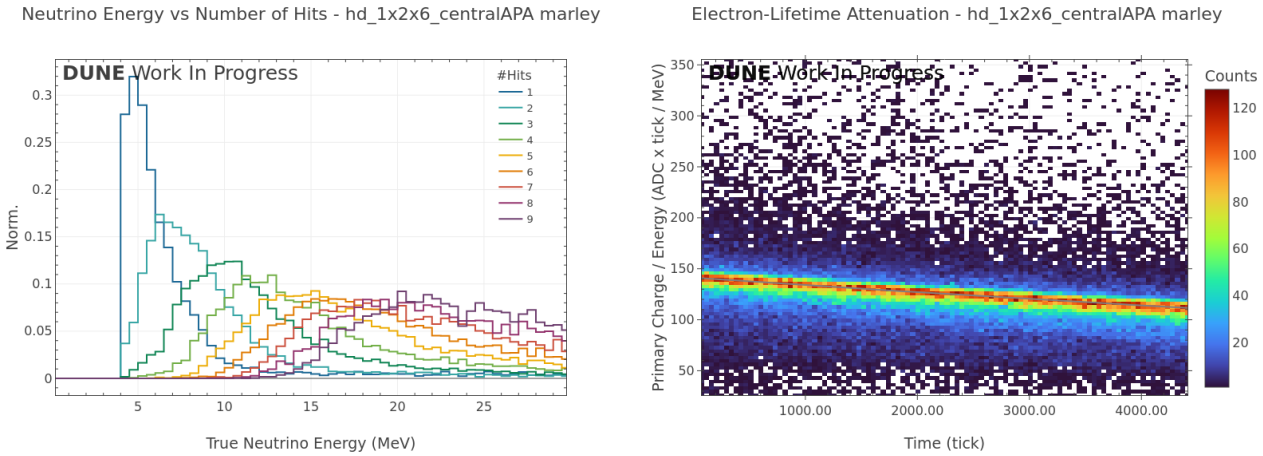


Figure 9: (left) Neutrino energy correlation with number of hits in the primary cluster. (right) Drift attenuation for the main electron cluster.

$$\frac{Q_{\text{raw}}(n_{\text{hit}})}{E_e^{\text{truth}}} = F_0(n_{\text{hit}}) \cdot \exp\left(\frac{t^{\text{drift}}}{\tau_e}\right), \tag{2}$$

where Q_{raw} is the observed (uncorrected) cluster charge, E_e^{truth} is the true main electron energy of the interaction, F_0 is the charge–to–energy correction factor, t^{drift} is the true drift time of the electron, and τ_e is the electron lifetime measured here independently and cross-checked against the simulation input.

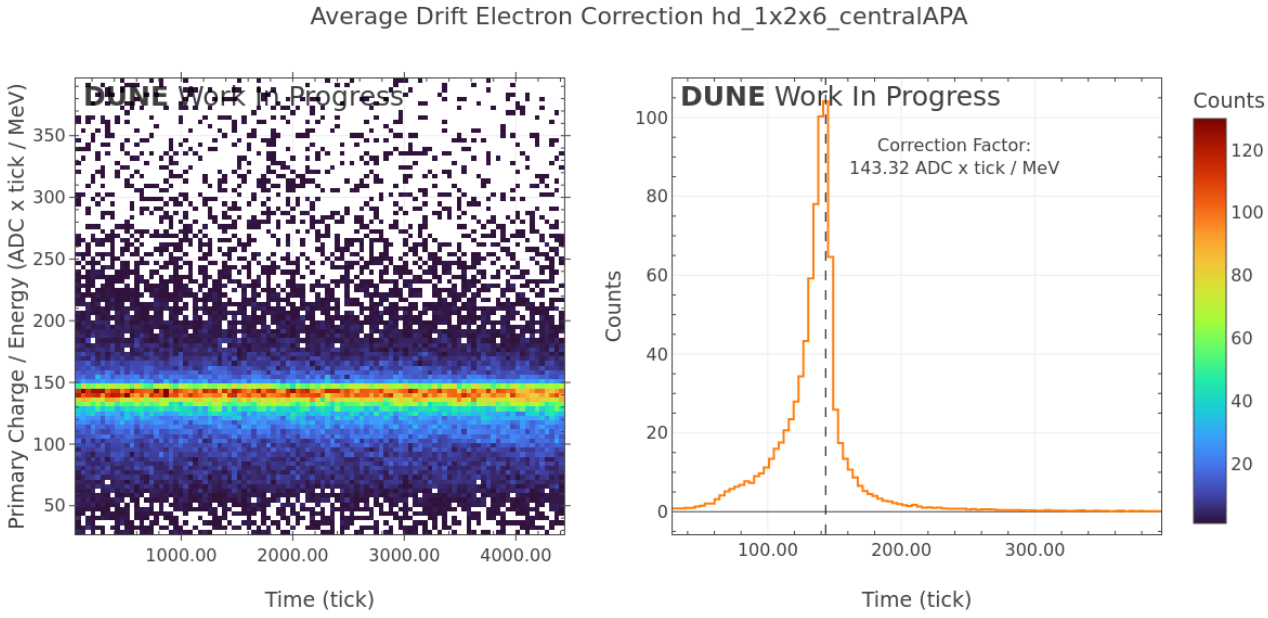


Figure 10: Drift attenuation correction.

Assuming the main cluster corresponds to a fully-contained electron, we take its known true energy (from simulation) as a reference and compute the charge–to–energy conversion factor for each (n_{hit}) threshold. Addi-

tionally, to assess the impact of this assumption, with the goal of characterizing signal clusters in the context of background which limits the clustering parameters, one can also investigate a "cheated charge". This is computed for all clusters, adding the surrounding charge that has been backtracked to the main electron of the interaction. A calibration based on this cheated clustering gives us an understanding of the losses that arise from a limited clustering ability in the context of background.

$$E_{\text{cluster}} = \alpha(n_{\text{hit}}, E_{\text{true}}) \cdot Q_{\text{corr}}. \tag{3}$$

After the truth drift correction has been applied to the data, the charge correction factor is immediately obtained.

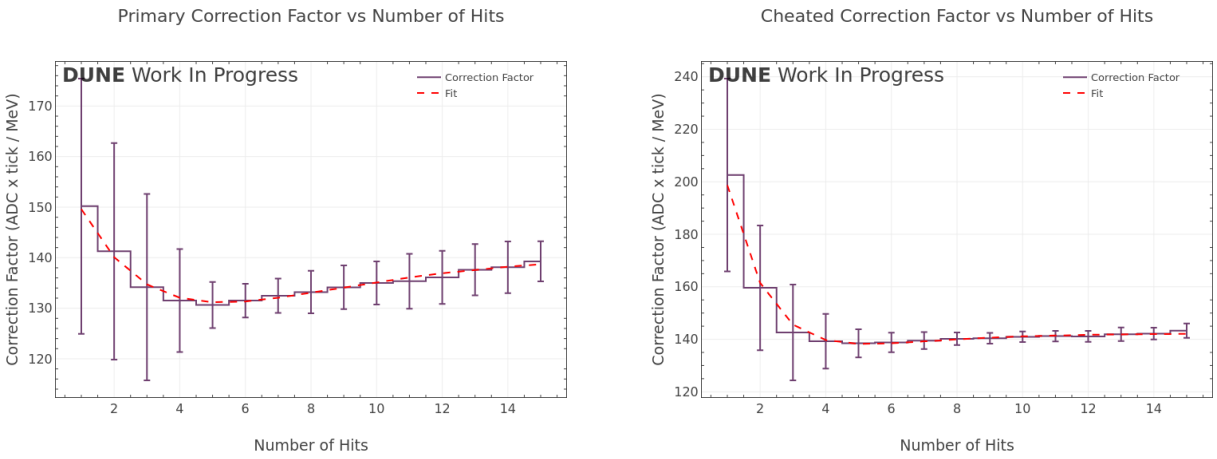


Figure 11: Per hit correction factor distributions. (left) Primary cluster charge. (right) Cheated electron cluster charge.

3.4 Electron Energy Calibration

Even after lifetime and charge calibration, small systematic biases may remain in the reconstructed electron energy due to recombination modelling, imperfect response, or clustering inefficiencies. We apply a linear correction to account for this residual offset:

$$E_e^{\text{final}} = (E_{\text{cluster}} - n)/m, \tag{4}$$

where m and n are derived from the linear fits to the simulated E_{true} versus E_{reco} distributions as shown in fig. 12. The calibration is performed separately for the main electron cluster and the cheated electron cluster. The calibration factor is computed as a function of the number of hits in the cluster (see fig. 13), which serves as a proxy for the electron energy.

3.5 Electron Energy Resolution

As a figure of merit for the process, we can compute the energy resolution from the cheated calibrated energy of the main electron. This is typically the value that analyzers compute when estimating the energy resolution of a TPC. It is obtained for an idealized clustering with ad hoc calorimetry and a high hit threshold (in this case we use 3 for a typical track). The energy resolution is computed as the Root Mean Square (RMS) of the relative error of the reconstructed variable versus the truth. The results are shown in fig. 14. The formula for the energy resolution fit is:

$$\Omega_{(E_{\text{true}})} = \sqrt{\left(\frac{p_2}{\sqrt{E_{\text{true}} - b}}\right)^2 + \left(\frac{p_1}{E_{\text{true}} - b}\right)^2 + p_0^2}, \tag{5}$$

where p_0, p_1, and p_2 are the parameters of the fit, and b is a constant offset and corresponds to the detection threshold. These parameters can be interpreted as the intrinsic resolution evaluated to 11(1)%, the stochastic term, and the noise term, respectively.

Applying this calibration to the main electron cluster, we obtain the final calibrated energy of the main electron. For all the adjacent clusters, we simply apply the charge correction factor obtained from the main electron cluster, assuming that these clusters are gamma-induced "blips" and therefore have a similar response to the main electron. fig. 15 shows the smearing matrices of the main electron and gamma energies after the

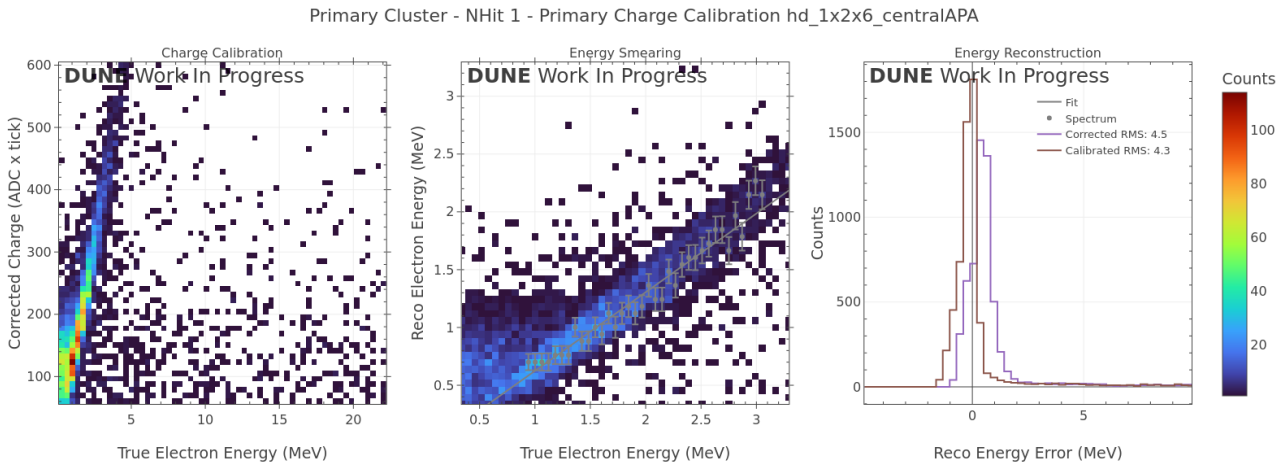


Figure 12: Charge calibration for the main electron cluster. (left) Corrected charge over true electron energy. (center) Linear fit to reco energy over true electron energy. (rigth) Comparison between (true - reco) for corrected (before linear fit calibration) and calibrated (after linear fit calibration) reco energy distributions.

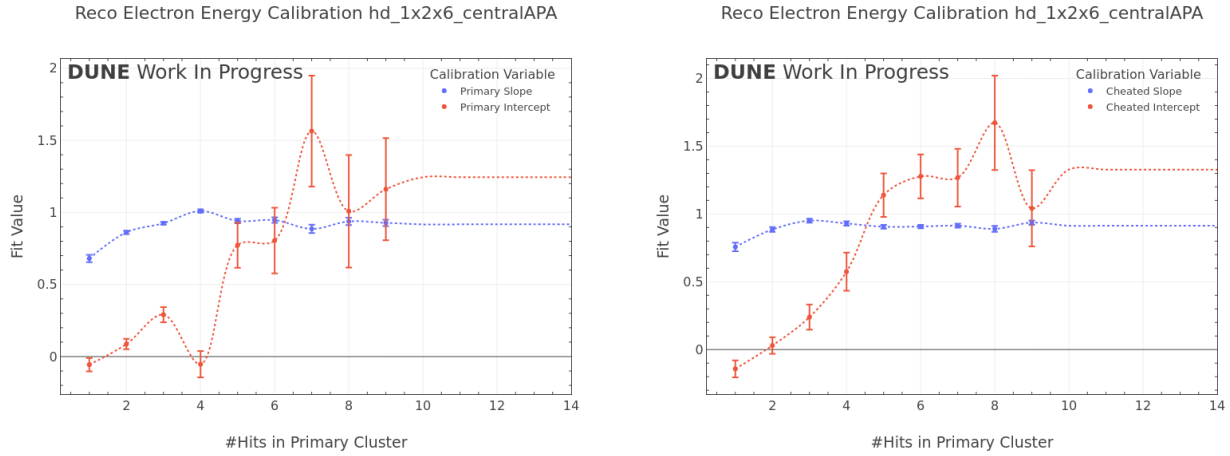


Figure 13: Per hit calibration results. (left) Primary cluster charge. (right) Cheated electron cluster charge. Blue shows the slope and red the intercept values of the linear fits.

calibration. Additionally, the smearing matrix of the visible energy (defined as the sum of the main electron and gamma energy) is shown.

3.6 Neutrino Energy Calibration

The main electron energy is corrected for the drift time attenuation and the charge calibration factor, as described in section 3.4. The main electron energy is then used to compute the total neutrino energy, which is computed from the main electron energy and the energies of all adjacent clusters (up to 1 m). The adjacent clusters are assumed to be gamma-induced "blips" and are calibrated using the same charge calibration factor as the main electron. The total neutrino energy is computed as:

$$E_\nu^{\text{raw}} = E_e^{\text{final}} + \sum_i E_{\gamma,i}^{\text{final}}. \tag{6}$$

This strategy is the first and most obvious approach for neutrino energy reconstruction (**TotalEnergy**). However, it does not account for the energy smearing of the signal, and the presence of background can significantly affect the reconstructed neutrino energy. Therefore, we study alternative methods to improve the energy resolution and mitigate the effects of background contamination.

The next simple solution is to apply a selection on the adjacent clusters (**SelectedEnergy**), which are assumed to be gamma-induced "blips". The selection is based on the adjacent cluster charge and distance to the main electron cluster. The adjacent clusters are selected if they have a charge above a certain threshold and are within a certain distance to the main electron cluster. The thresholds are tuned to maximize the signal over

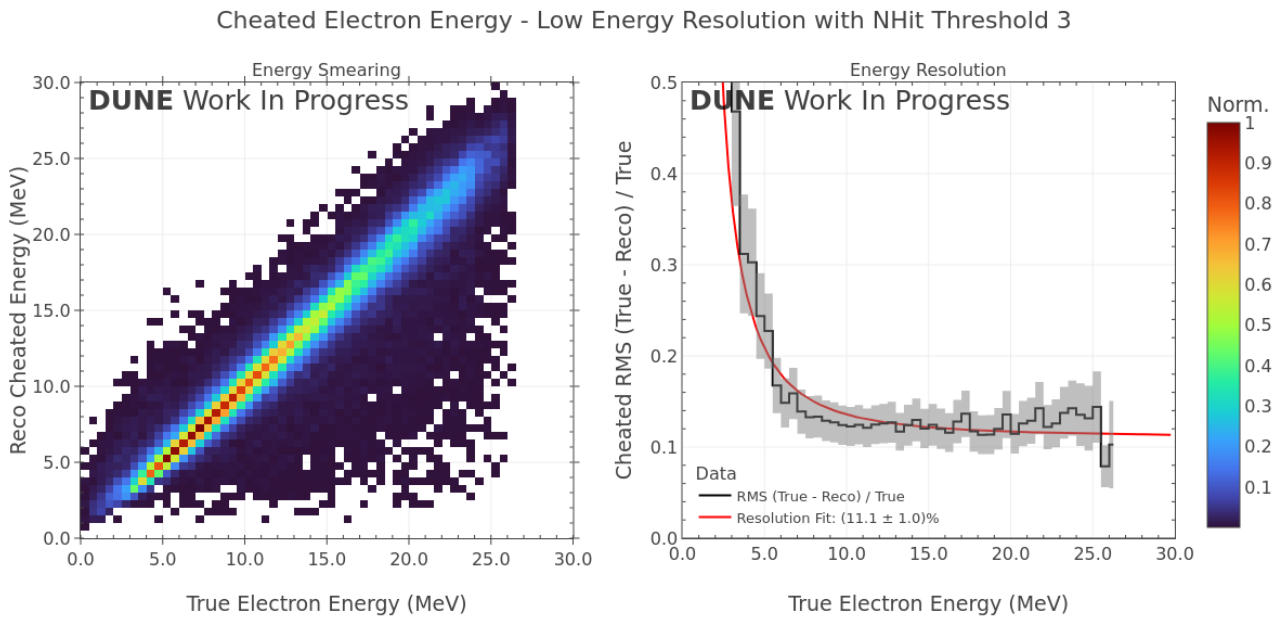


Figure 14: Energy resolution from calibrated charge with cheated clustering and hit threshold of 3.

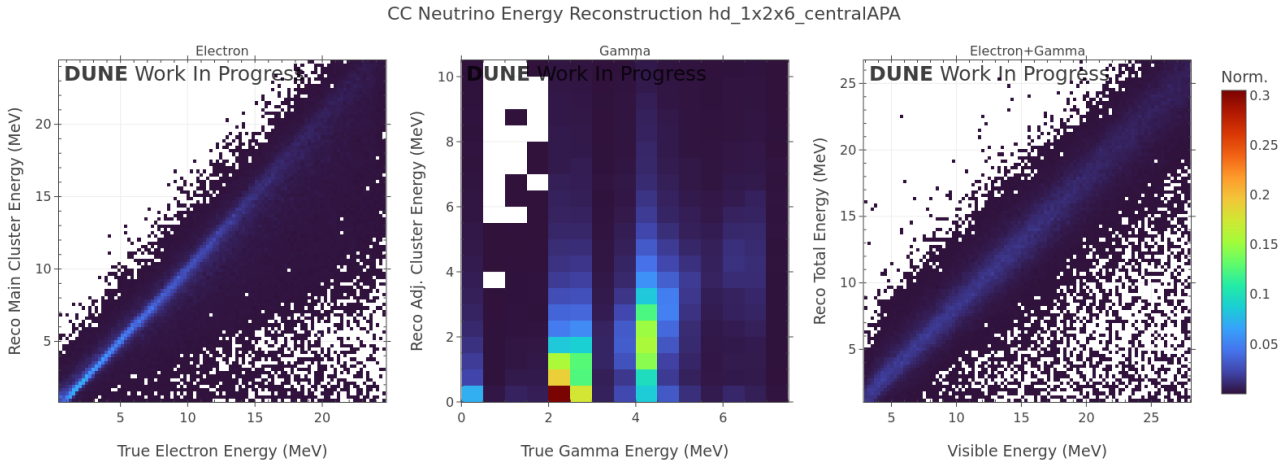


Figure 15: Signal daughter smearing obtained after main electron calibration.

background ratio. fig. 16 shows the distribution of the adjacent clusters charge and distance to the main electron cluster and selection thresholds.

The next step is to apply a machine learning algorithm to the adjacent clusters to improve the energy resolution. The algorithm is trained on the main cluster properties and the average adjacent cluster features. The algorithm is a random forest regressor, which is trained on the simulated data. The random forest regressor is used to predict the energy of the adjacent clusters, which are then added to the main electron energy to obtain the total neutrino energy (**SolarEnergy**). To simply the problem and motivated by the two main calibration lines from the CC interaction displayed in fig. 6, the Machine Learning (ML) algorithm is trained two add one of two total adjacent cluster energy values: 4.47 MeV and 6.41 MeV. fig. 17 shows the score of the random forest regressor which provides an AUC of 0.77 for the test sample, which means that the algorithm is able to distinguish between signal and background with a good efficiency. The random forest regressor is trained on the following features:

- NHits: number of hits in the main electron cluster.
- AdjClNum: number of total adjacent clusters.
- SelectedAdjClNum: number of selected adjacent clusters.
- Energy: energy of the main clusters.

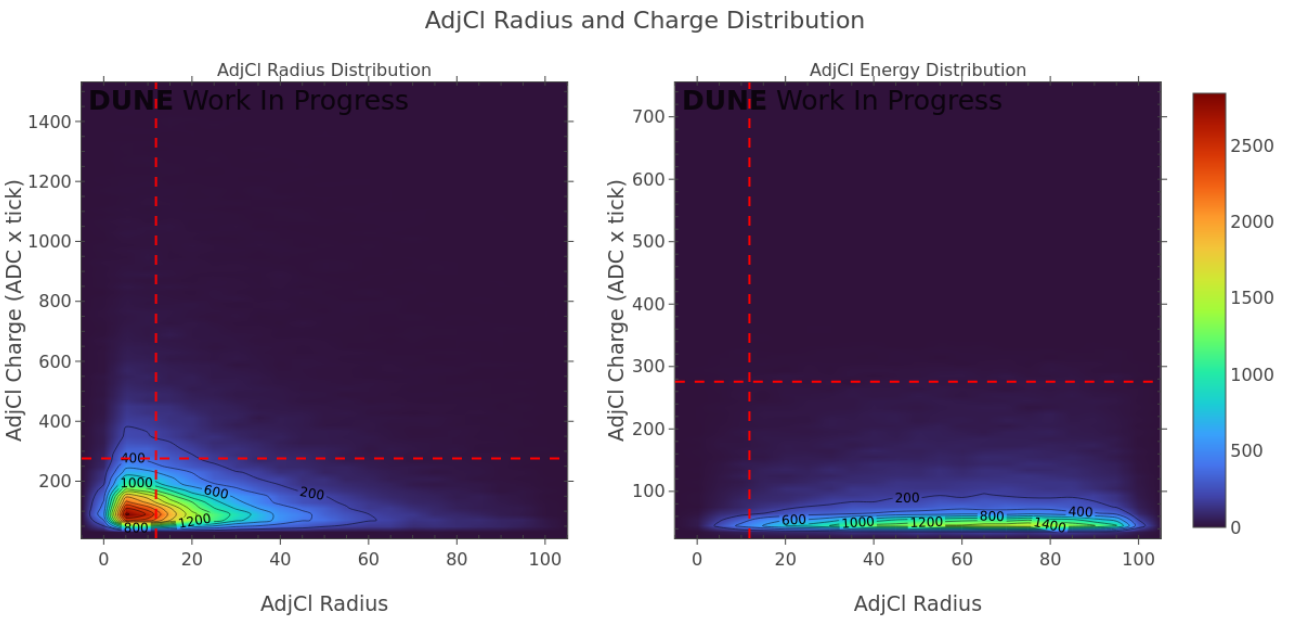


Figure 16: Distribution of the adjacent clusters charge and distance to the main electron cluster.

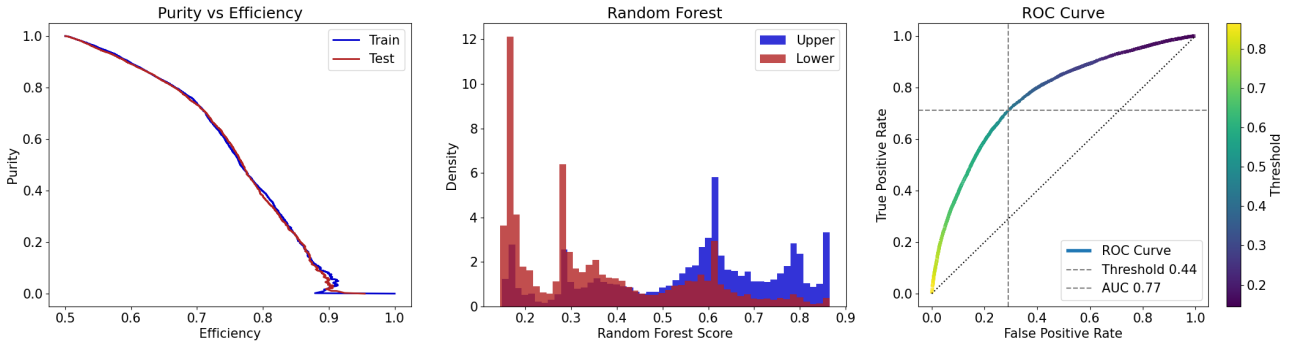


Figure 17: (left) Purity vs Efficiency curve of the random forest regressor for both the train and the test samples. (center) Score of the random forest regressor for both CC samples. (right) ROC curve of the random forest regressor.

- **TotalAdjClEnergy**: energy of the total adjacent clusters.
- SelectedAdjClEnergy: energy of the selected adjacent clusters.
- SelectedAdjClEnergyRatio: energy ratio of the selected to the total adjacent clusters.
- **MaxAdjClEnergy**: maximum energy of the total adjacent clusters.
- SelectedMaxAdjClEnergy: maximum energy of the selected adjacent clusters.
- TotalEnergy: reconstructed **TotalEnergy**.
- SelectedEnergy: reconstructed **SelectedEnergy**.

From which the highlighted features are the most important for the random forest regressor, as shown in fig. 18.

Finally, as a benchmark for comparison between the here presented reconstruction algorithms and with the aim of quantizing the impact of adjacent gammas on the total neutrino energy reconstruction, we introduce the **ClusterEnergy** algorithm as the reconstruction method based purely on the main electron cluster. The only correction is a fixed calibration offset applied to the full sample to align the center of the distribution with the true energy.

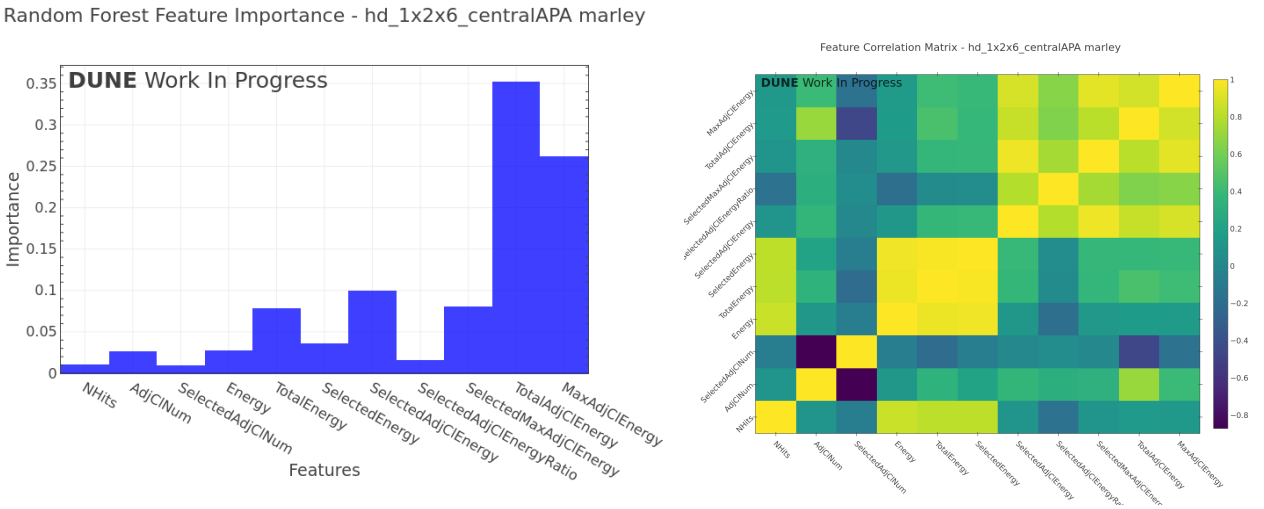


Figure 18: (left) Feature importance of the random forest regressor. (right) Correlation matrix of the features.

3.7 Neutrino Energy Smearing

Due to the missing nuclear binding energy and undetected soft particles, the reconstructed neutrino energy is expected to systematically underestimate the true value. As a final step to the reconstruction process, all energies (**TotalEnergy**, **SelectedEnergy**, **SolarEnergy** and **ClusterEnergy**) are calibrated with a linear fit to compensate for systematic losses in the previous clustering or calibration processes. This final global correction is applied to bring the distribution into agreement with the simulation truth:

$$E_\nu^{\text{cal}} = \left(E_\nu^{\text{raw}} - n\right)/m, \tag{7}$$

with m and n obtained from a linear fit to the true versus reconstructed neutrino energy spectrum of the MC sample as shown in fig. 19.

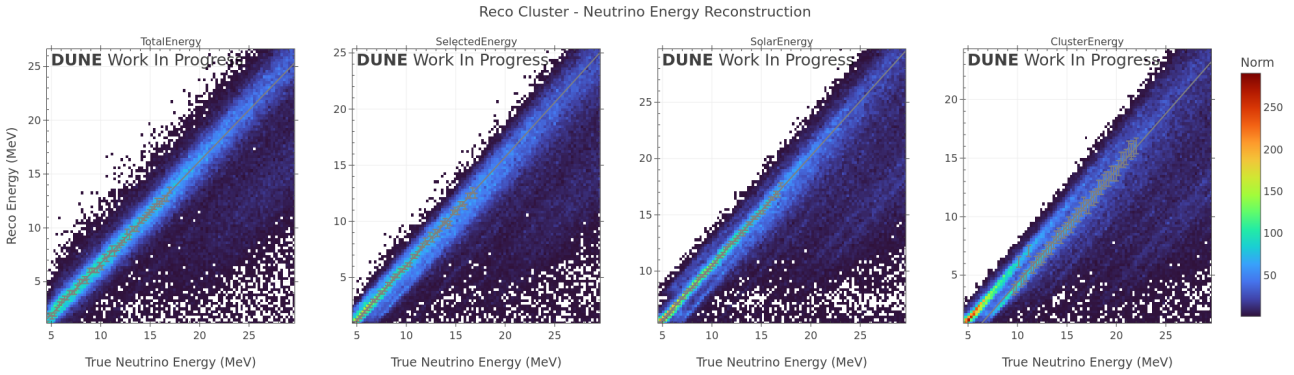


Figure 19: Neutrino energy smearing from the calibrated signal.

3.8 Flash Matching

The final step in the signal reconstruction process is to match the reconstructed TPC clusters with the PDS flashes. The matching is performed by evaluating the distance between the TPC vertex and the PDS flash position, as well as the time difference between the TPC cluster and the PDS flash. The matching efficiency is defined as the fraction of TPC clusters that have a corresponding pure PDS flash. fig. 20 shows the flash–matching efficiency as a function of the distance to the PDS plane and the neutrino energy. The matching efficiency is high for distances up to 1 m and decreases for larger distances. The matching efficiency is also higher for higher neutrino energies. This is expected, as the PDS flashes are more likely to be detected for higher energy depositions.

For any given TPC cluster, the matching strategy follows these steps:

1. **Distance Calculation**: Compute the distance between the TPC cluster vertex and all PDS flash positions.

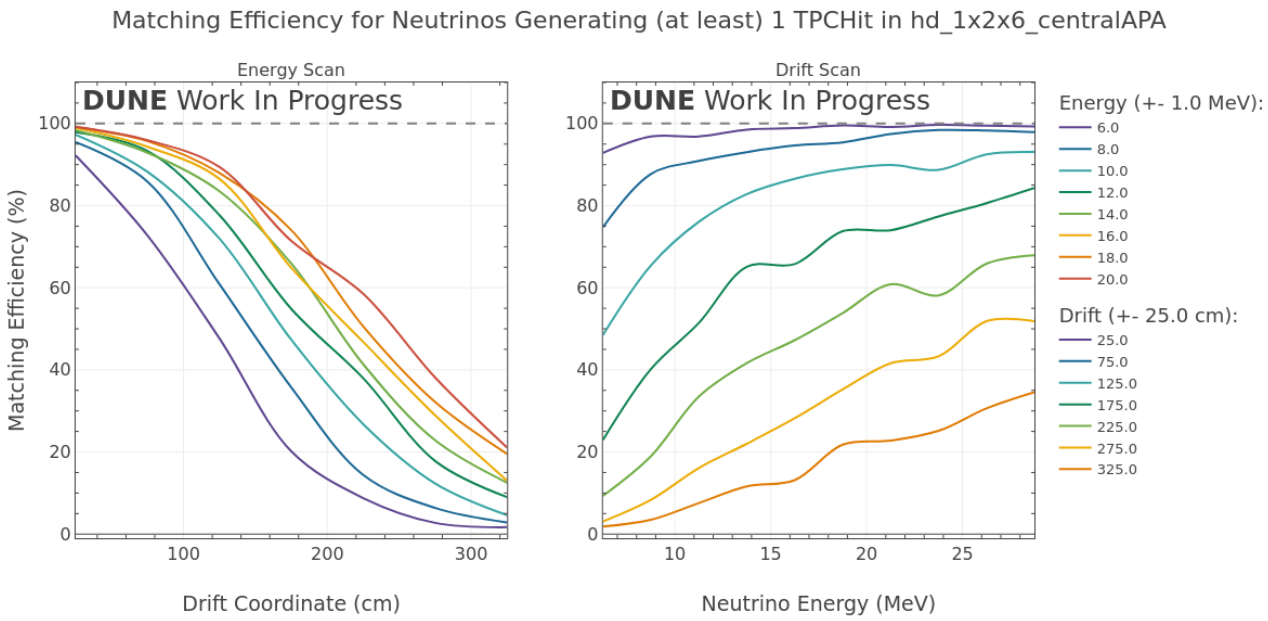


Figure 20: Flash–matching efficiency. (left) As a function of distance to the PDS plane. (right) As a function of neutrino energy.

2. **Time Difference Calculation**: Calculate the time difference between the TPC cluster and the PDS flash. This must be smaller than the drift time of the electrons in the LAr.
3. **Candidate Selection Criteria**: A PDS flash is considered adjacent to a TPC flash if the distance $(\mathrm{TPC_y} - \mathrm{PDS_y})^2/\mathrm{AdjOpFlashY}^2 + (\mathrm{TPC_Z} - \mathrm{PDS_Z})^2/\mathrm{AdjOpFlashZ}$ is smaller than 1, where AdjOpFlashY and AdjOpFlashZ are the adjusted matching distances in the 2 projected coordinates, respectively.
4. **Matching Criteria**: The algorithm consider 3 additional cuts to prevent the random matching of background cluster:
   - The minimum number of hits in the PDS flash: AdjOpFlashMinNHitCut (by default 3).
   - The minimum number of PEs in the PDS flash: AdjOpFlashMinPECut (by default 20).
   - The maximum ratio of AdjOpFlashMaxPE / AdjOpFlashPE: AdjOpFlashMaxPERatioCut (by default 1). This is intended to prevent the matching of background PDS flashes, wich are less homogeneous in the number of PEs compared to the signal PDS flashes.

Select the PDS flash with the highest number of PE as the matching candidate for the TPC cluster. If no PDS flash is found within the matching criteria, the TPC cluster is considered unmatched.

From this matching one can reconstruct the drift coordinate of the TPC cluster, which is essential for the correct reconstruction of the neutrino energy. The drift coordinate is computed from the time between the TPC vertex and the PDS flash by taking into account the drift velocity of the electrons in the LAr.

4 Results

From the TPC cluster vertex reconstruction and the flash–matching, we can extract the energy and vertex resolution. The calibrated neutrino energy is obtained from a standalone code SOLAR package, which formats the simulation output into numpy arrays and performs the successive operations presented in section 3. Here we present the main figures of merit and compare the resolution performance between the two geometries, the hd_1x2x6_centralAPA and the vd_1x8x14_3view_30deg and additional background reduced scenarios.

4.1 Neutrino Energy Resolution

The energy resolution is defined as the ratio of the standard deviation of the reconstructed neutrino energy to the mean value of the neutrino energy. The energy resolution is calculated from the calibrated neutrino energies. The results are shown in fig. 21 for the 4 different reconstruction algorithms.

As expected, the **ClusterEnergy** algorithm has the worst energy resolution for neutrino energies below 10 MeV, as it does not take into account the energy deposited by adjacent gamma blips which can amount to

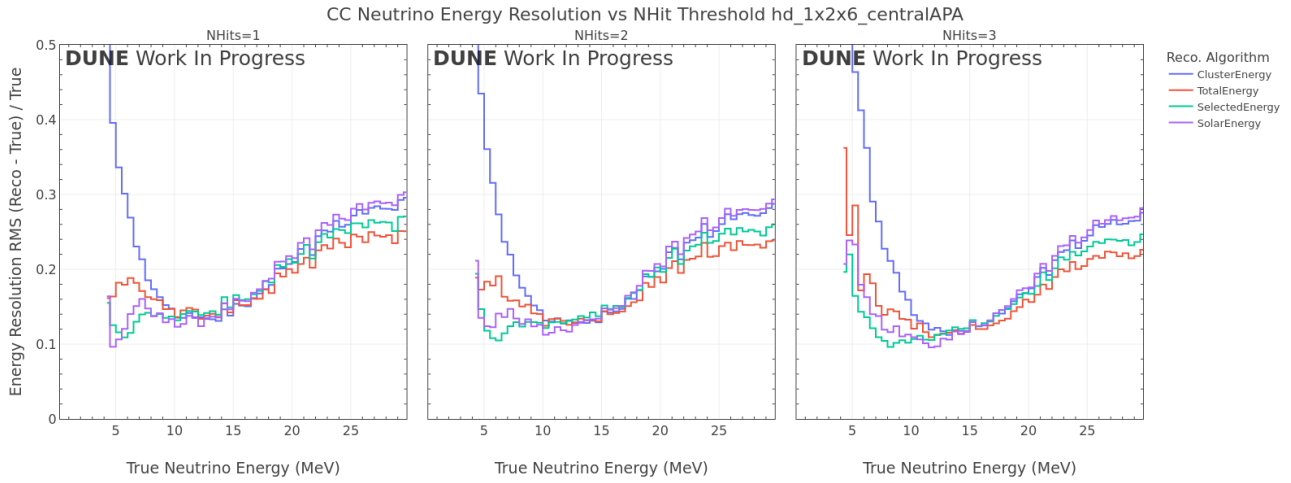


Figure 21: Energy resolution from calibrated neutrino energies with different reconstruction algorithms.

more than 50% of the initial neutrino energy. The **SelectedEnergy** and **SolarEnergy** algorithms have the best resolution in this range. As the energy of the neutrino goes higher, the relative weight of the adjacent background clusters starts becoming less important. Over the range of 10 MeV and 20 MeV all algorithms perform almost identical and for the 20 MeV and 30 MeV neutrino energies, the **TotalEnergy** algorithm starts to outperform the others. Importantly, this trend is maintained for any hit threshold selected, ensuring that the energy resolution is robust against the clustering criteria.

Overall, the **SolarEnergy** algorithm is the best choice for neutrino energies below 20 MeV, as it has better (or compatible) energy resolution compared to the other algorithms. The benefit of this method that is not shown in this comparison is that it sets an upper limit to the reconstructed energies of background particles. If not constrained, cavern backgrounds such as neutrons or gammas can contribute significantly with fake events to physics analyses such as the solar neutrino oscillation studies. By only adding a maximum fixed amount of energy to the main clusters produced by these particles, one ensures that their contribution is limited to lower energies compared to the full scope of the solar neutrino spectrum.

4.2 Neutrino Vertex Resolution

The vertex resolution is defined from the distance between the reconstructed vertex and the true vertex position.

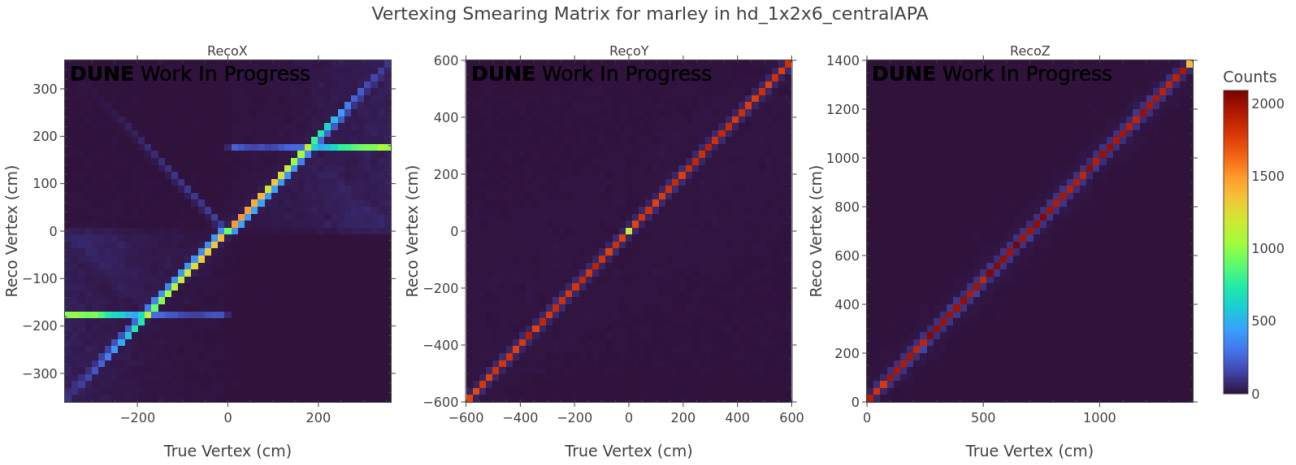


Figure 22: Vertex smearing from calibrated signal.

The vertex resolution is calculated from the calibrated signal, which is obtained from the TPC cluster vertex reconstruction. The results are shown in fig. 23. Notice how the z–axis of the x-coordinate heatmaps shows fewer counts than the rest. This is a result of the flash–matching inefficiencies.

Form this computation we obtain a vertex resolution that is quite poor. As a figure of merit, the resolution computed in this way behaves correctly if the reconstructed values are normally distributed around the truth. This is not the case of the vertex reconstruction algorithms in a LArTPC where failures in the TPC induction–

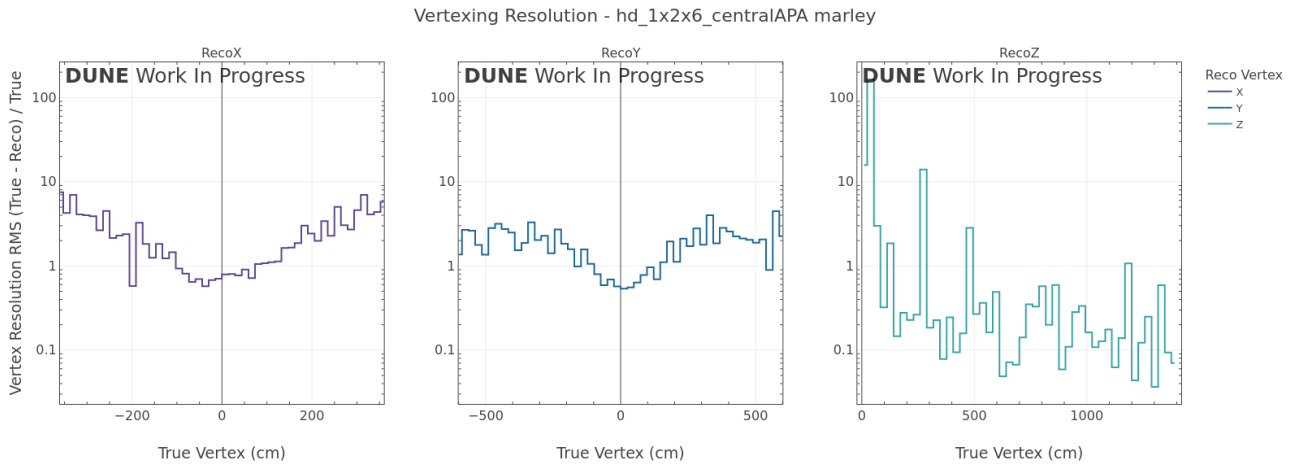


Figure 23: Vertex resolution from calibrated signal.

collection plane or the flash–matching algorithms lead to arbitrary values. TO have a better understanding of the reconstruction potential, we can compute the cumulative vertex error fig. 24.

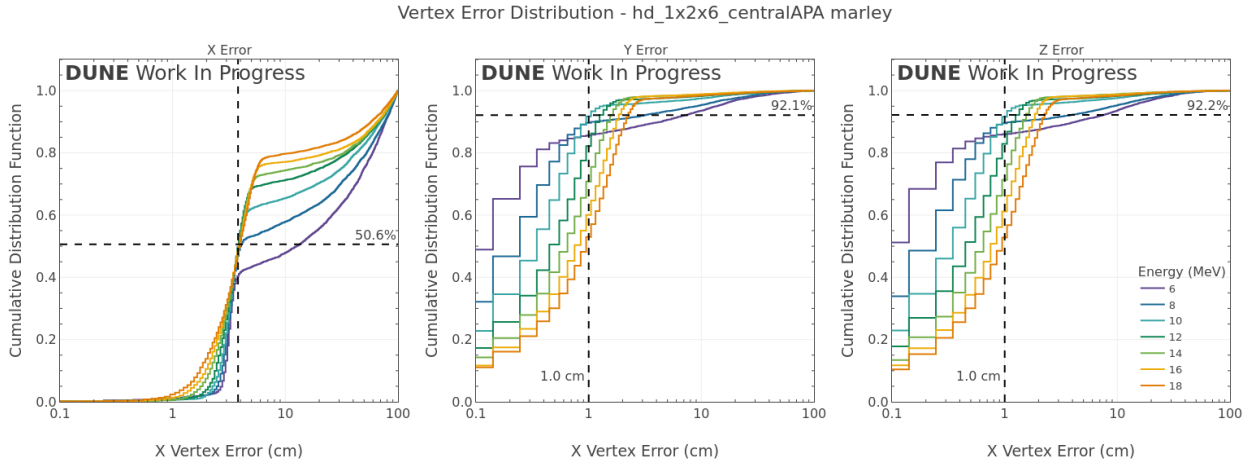


Figure 24: Cumulative vertex error from calibrated signal.

As a figure of merit to extract from this analysis, we can find the point that minimizes the second derivative of these curves. In terms of reconstruction this corresponds to the inflexed point between the events that have been correctly matched and the ones that have a pseudo–random reconstruction. By doing so we find the vertex tolerance, and we evaluate the percentage of the sample that is reconstructed within that value. By doing so, we observe better results in the y– and z–coordinates. This is expected since the x–coordinate is obtained from the flash–matching algorithm, which has a higher inefficiency than the TPC cluster vertex reconstruction.

4.3 DUNE Geometry Comparison

So far we have used as reference the *hd_1x2x6_centralAPA* geometry, corresponds to the inner drift volumes of the HD detector. In this section, we compare the results with the *vd_1x8x14_3view_30deg* geometry as well as different background scenarios (nominal, optimistic and shielded), which are described in section 2.1.

Applying the **TotalEnergy** reconstruction algorithm, we can see that the energy resolution is better in the hd_1x2x6_centralAPA geometry, which is expected since it has a lower background rate and smaller drift volume causing therefore less diffusion. The energy resolution is shown in fig. 25.

Applying the **SolarEnergy** reconstruction algorithm we can mitigate this difference in flavour of the vd_1x8x14_3view_30de geometry, which has a lower TPC threshold and is able to pitch up more resolution on the primary cluster. The energy resolution is shown in fig. 26.

In the case of the vertex resolution, we can use the previously defined figure of merit and quote the percent of events with a vertex error below a certain threshold. The results are shown in table 1.

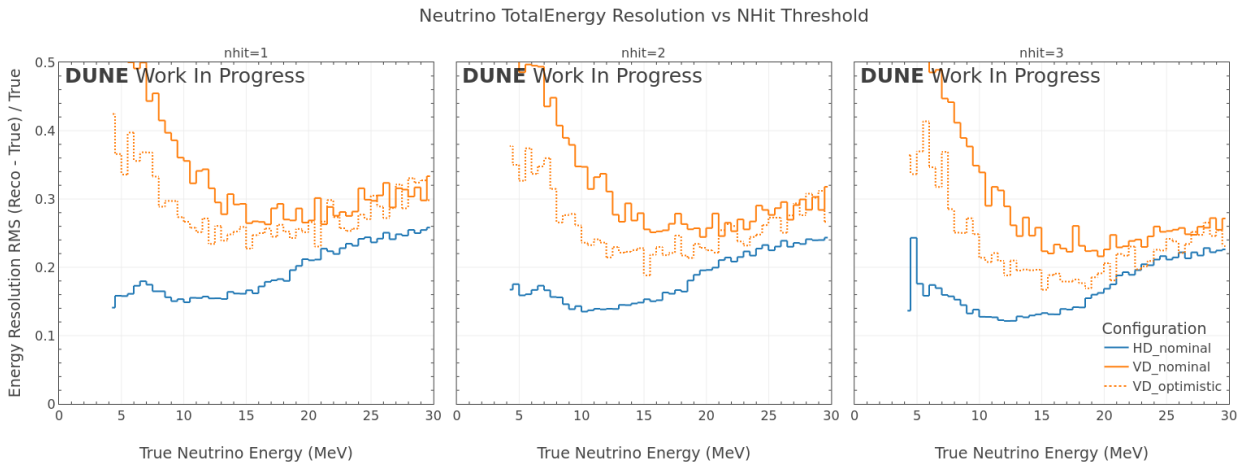


Figure 25: Energy resolution comparison.

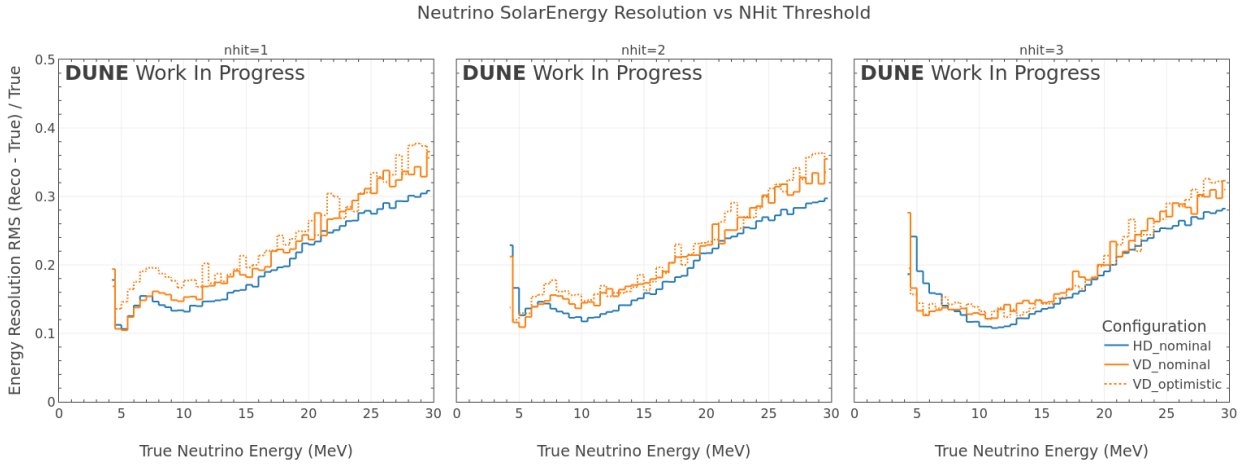


Figure 26: Energy resolution comparison.

4.4 Flash–Matching Effects on Resolution

As already explained, the flash–matching step is a crucial in the reconstruction process, as it allows to associate a TPC clusters with the corresponding flash and hence obtain the correct drift coordinate of the interaction.

Add flash–matching effects on the energy resolution.

5 Conclusion

In this technical note, we have presented the reconstruction algorithms for the DUNE FD, focusing on the neutrino energy and vertex resolution. The reconstruction is based on the TPC cluster vertex reconstruction and flash–matching, which allows us to extract the calibrated neutrino energy and vertex position.

	Tolerance – Sample Size Below Tolerance (%)		
Geometry	**X**	**Y**	**Z**
hd_1x2x6_centralAPA	3.9 cm – 50.6%	1 cm – 92.1%	1 cm – 92.2%
vd_1x8x14_3view_30deg	1 cm – 15.3%	1 cm – 89.5%	1 cm – 90.5%
vd_1x8x14_3view_30deg_optimistic			

Table 1: Vertex resolution comparison between geometries. Percentage of 10 MeV sample below coordinate tolerance.

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List of Acronyms

APA Anode Plane Assembly

CC Charged Current

CPA Cathode Plane Assembly

DUNE Deep Underground Neutrino Experiment

ES Elastic Scattering

FD Far Detector

HD Horizontal–Drift

LAr Liquid Argon

LArSoft Liquid Argon Software Framework

LArTPC Liquid Argon Time Projection Chamber

MARLEY Model of Argon Reaction Low Energy Yields

MC Monte Carlo

ML Machine Learning

NC Neutral Current

PDS Photon Detection System

PE Photoelectron

RMS Root Mean Square

PMT Photomultiplier Tube

TPC Time Projection Chamber

VD Vertical–Drift

XA X–ARAPUCA